CLASS 10 MATHEMATICS :: SLIP TEST – FA-4

Maximum Marks: 20 • Time: 45 Minutes

Name: _____ **Roll No:** _____ **Sec:** _____ **Marks:** _____

SECTION I: Short Answer Questions — 3 × 2M = 6 Marks

1. If $\tan\theta = 4/3$, find the values of $\sin\theta$ and $\cos\theta$ for acute angle θ. **[2 M]**

2. A box contains 5 red balls, 8 white balls, and 4 green balls. One ball is taken out at random. Find the probability that the ball drawn is (i) white and (ii) not green. **[2 M]**

3. Find the volume of a right circular cylinder whose base radius is 7 cm and height is 10 cm. (Take $\pi = 22/7$) **[2 M]**

SECTION II: Long Answer Questions — 2 × 4M = 8 Marks

4. A 1.5 m tall observer is 28.5 m away from a tower. The angle of elevation of the top of the tower from his eyes is 45°. Find the height of the tower. **[4 M]**

5. A solid toy is in the form of a cylinder surmounted by a hemisphere of the same radius. The common radius is 7 cm and the total height of the toy is 17 cm. Find: **[4 M]**

(i) The curved surface area of the cylinder ($2\pi rh$).

(ii) The curved surface area of the hemisphere ($2\pi r^2$).

(iii) The total volume of the combined solid. (Take $\pi = 22/7$)

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Multiple Choice Section

SECTION III: Multiple Choice Questions **6 × 1M = 6 Marks**

6. The value of (sin 30° + cos 60°) − tan 45° is:

(A) 1 (B) 0 (C) −1 (D) 1/2

7. The trigonometric identity $1 + \tan^2 \theta$ is identically equal to:

(A) $\sec^2 \theta$ (B) $\operatorname{cosec}^2 \theta$ (C) $\cot^2 \theta$ (D) $\cos^2 \theta$

8. If the angle of elevation of the sun is 45°, the length of the shadow of a 15 m high pole is:

(A) $15\sqrt{3}$ m (B) 15 m (C) $15/\sqrt{3}$ m (D) 30 m

9. Which of the following cannot be the probability of an event?

(A) 2/3 (B) −1.5 (C) 15% (D) 0.7

10. Two unbiased coins are tossed together. The probability of getting at most one head is:

(A) 1/4 (B) 1/2 (C) 3/4 (D) 1

11. A metallic sphere of radius 3 cm is melted and recast into small spheres of radius 1 cm. The number of small spheres formed is:

(A) 9 (B) 18 (C) 27 (D) 36

CLASS 10 MATHEMATICS :: SOLUTIONS – FA-4

Section I & Section II Solutions

SECTION I: Short Answer Solutions — 6 Marks

Solution 1 (Trigonometric Ratios) — 2 M

tan θ = 4/3 = Opposite / Adjacent

Hypotenuse = $\sqrt{[(4)^2 + (3)^2]} = \sqrt{(16 + 9)} = \sqrt{25} = 5$

sin θ = Opposite / Hypotenuse = 4/5

cos θ = Adjacent / Hypotenuse = 3/5

sin θ = 4/5 | cos θ = 3/5

Solution 2 (Probability) — 2 M

Total balls = 5 (red) + 8 (white) + 4 (green) = 17

(i) P(White) = Favourable / Total = 8/17

(ii) P(Not green) = (5 + 8) / 17 = 13/17

P(White) = 8/17 | P(Not green) = 13/17

Solution 3 (Cylinder Volume) — 2 M

r = 7 cm, h = 10 cm

$V = \pi r^2 h = (22/7) \times 7 \times 7 \times 10$

$= 22 \times 7 \times 10 = 154 \times 10 = 1540\ cm^3$

Volume = 1,540 cm^3

SECTION II: Long Answer Solutions — 8 Marks

Solution 4 (Height of Tower) — 4 M

Observer height h_1 = 1.5 m, Distance d = 28.5 m, Angle of elevation θ = 45°

Let top segment of tower = h_2

In right Δ: tan 45° = h_2 / d

1 = h_2 / 28.5 ➔ h_2 = 28.5 m

Total height of tower H = $h_1 + h_2$ = 1.5 + 28.5 = 30.0 m

Height of the tower = 30 m

Solution 5 (Combined Solid) — 4 M

Common radius r = 7 cm, Total height = 17 cm ➔ Height of cylinder h = 17 − 7 = 10 cm

(i) CSA of Cylinder: $2\pi rh = 2 \times (22/7) \times 7 \times 10 = 440\ cm^2$

(ii) CSA of Hemisphere: $2\pi r^2 = 2 \times (22/7) \times 7 \times 7 = 308\ cm^2$

(iii) Total Volume: $V_{cyl} + V_{hemi} = \pi r^2 h + (2/3)\pi r^3$

$V_{cyl} = (22/7) \times 7 \times 7 \times 10 = 1540\ cm^3$

$V_{hemi} = (2/3) \times (22/7) \times 7 \times 7 \times 7 = 2156 / 3 = 718.67\ cm^3$

Total Volume = 1540 + 718.67 = 2258.67 cm^3

Answers: (i) 440 cm^2 | (ii) 308 cm^2 | (iii) 2,258.67 cm^3

CLASS 10 MATHEMATICS :: SOLUTIONS – FA-4

Section III Multiple Choice Solutions

SECTION III: Multiple Choice Solutions — **6 Marks**

Solution 6 — **1 M**

$(\sin 30° + \cos 60°) - \tan 45°$

$= (1/2 + 1/2) - 1 = 1 - 1 = 0$

Answer: (B) 0

Solution 7 — **1 M**

Standard identity: $1 + \tan^2 \theta = \sec^2 \theta$

Answer: (A) $\sec^2 \theta$

Solution 8 — **1 M**

$\tan 45° =$ Height / Shadow ➔ $1 = 15$ / Shadow

Shadow = 15 m

Answer: (B) 15 m

Solution 9 — **1 M**

For any event E: $0 \le P(E) \le 1$. Probability cannot be negative.

Answer: (B) −1.5

Solution 10 — **1 M**

Outcomes: {HH, HT, TH, TT} ➔ Total = 4

At most one head: {HT, TH, TT} ➔ 3 favourable

P = 3/4

Answer: (C) 3/4

Solution 11 — **1 M**

N = (Volume of big sphere) / (Volume of small sphere)

$= (4/3\ \pi R^3) / (4/3\ \pi r^3) = (3/1)^3 = 27$

Answer: (C) 27